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Review

Transdermal Uptake of Substances Banned in Sports and Its Relevance for Doping Controls

Sophia Krombholz[1], Linus Korsmeier[1], Mario Thevis[2]
Affiliation addresses are listed at the end of the article.

ABSTRACT

Transdermal drug delivery has emerged as an area of significant interest in both pharmaceutical sciences and antidoping research. This narrative review explores the implications of transdermal uptake of substances banned in sports, emphasizing its relevance for doping control analysis. The human skin's complex role as both a barrier and a site of drug absorption is examined, alongside studies investigating the transdermal administration of substances on the World Anti-Doping Agency Prohibited List, including approved transdermal formulations and drug candidates. Recent cases demonstrate that (unintentional) dermal exposure of athletes to substances banned in professional sports, e.g., clostebol, can lead to an adverse analytical finding in doping controls. Consequently, studies investigating the detectability and metabolism of transdermally administered doping substances, which might even support identifying the route of exposure, are becoming increasingly important. This review emphasizes the relevance of addressing the skin as a pathway for both intentional doping and unintentional contamination, outlining critical implications for global antidoping efforts and future research priorities.

Keywords Sport, Doping, Exposure, Skin

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Correspondence Dr. Sophia Krombholz, Institute of Biochemistry, German Sport University Cologne, Am Sportpark Müngersdorf 6, 50933 Cologne, Germany, Email: s.krombholz@biochem.dshs-koeln.de

Introduction

For decades, transdermal drug administration has been explored as a way to improve the bioavailability of substances, simplify drug application, and/or minimize undesirable side effects.[1] This effort contrasts with the function of the skin as a barrier of the human body against environmental substances, and only substances exhibiting distinct physico-chemical properties penetrate the skin easily.[2,3] However, the desire to enable the transdermal delivery of substances that are otherwise hardly absorbed has led to a considerable amount of research and sophisticated strategies, to the extent that even proteins and vaccines can nowadays be administered via the skin.[4,5] In the context of sports doping, the percutaneous administration of prohibited substances is not a new concept either, since transdermal administration forms of the anabolic androgenic steroid (AAS) testosterone have been available since the 1980s.[6] Accordingly, the detection of transdermal testosterone in doping control analysis has been thoroughly investigated over the years.[7–9] More recently, however, the skin as a site of drug resorption has become relevant from a different perspective. Several cases of adverse analytical findings (AAFs) have been presented to the Court of Arbitration for Sport, in which athletes have attributed the presence of prohibited substances and/or their metabolites in their sample to unintentional and unknowing contamination via the skin.[10–12] A substance that was involved particularly often was the AAS clostebol, which is available for the treatment of minor skin injuries, e.g., in Italy (Trofodermin). The athletes argued exposure to clostebol while applying the cream to team colleagues, treating their dog, or during a massage by their physiotherapist, who had previously self-applied the cream.[10,13] Such a contamination scenario was simulated in a study by de la Torre et al., demonstrating that even very brief contact with the clostebol cream can result in findings of the AAS and its main metabolite in urine.[10] The possibility of transdermal absorption of clostebol and other synthetic steroids was also demonstrated by Gessner et al.; here, too, the parent substances and several metabolites were detected in the participants' urine samples.[14] The overall exposure of professional athletes to prohibited substances in their environment, for example, through their presence in food, dietary supplements, or intimate contact with another person, has previously been discussed as the "athletes' exposome."[15] Highly sensitive analytical methods covering a broad spectrum of chemicals and drugs paired with frequent routine testing significantly increase the risk of AAFs caused by contamination as opposed to deliberate doping.[15–17] In that regard, the skin has to be considered as a potential site of exposure, given its large surface that can be in contact with doping substances in its environment through various sources. This includes the—certainly dreaded—possibility of malicious sabotage, e.g., through a handshake, causing deliberate contamination of the athlete via the skin. Importantly, the athlete is held responsible for any prohibited substance found in his/her sample, regardless if it was ingested intentionally, unintentionally, or unknowingly.[18] Nonetheless, identifying potential contamination

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scenarios and, on the other hand, correctly determining antidoping rule violations (ADRVs) is vital for the integrity of the doping control system.[19] Thus, from a scientific point of view, studies on dose-dependent pharmacokinetics, especially those elucidating the biotransformation and excretion profile of transdermally administered doping substances, are important to support or question the plausibility of such claims. With regard to potential scenarios of contamination via the skin, analytical strategies that allow to distinguish between different routes of administration, e.g., oral and transdermal, could be particularly beneficial.[17] However, a plethora of factors such as age or skin condition can influence the absorption of substances through the skin and necessitate consideration.[20] In addition to the skin being a potential site of contamination, deliberate uptake of doping substances via the skin must be taken into account, as transdermal formulations can provide an easy means to administer performance-enhancing substances.[19] Furthermore, it has been suggested that transdermal delivery of anabolic steroids could be a way to circumvent positive findings in doping controls.[21] Therefore, investigating the urinary metabolite profiles and the detectability of substances administered via the transdermal route appears relevant and warranted.

Within this narrative review, the effect of transdermal administration on metabolism and elimination of compounds that are relevant in sports drug testing programs is discussed. The fundamental aspects of the human skin and transdermal drug delivery are summarized, and several factors are outlined that can influence skin absorption processes. Next, studies investigating the transdermal administration of several representative substances included in the World Anti-Doping Agency (WADA) Prohibited List are evaluated, including drugs approved as transdermal formulations, as well as substances currently subject to research studies. Finally, the influence of transdermal administration on the metabolism of doping substances is reviewed and the consequences and opportunities for antidoping research are evaluated.

A comprehensive literature search was conducted to identify relevant studies for this review, utilizing the electronic databases PubMed and Google Scholar. The search strategy incorporated a combination of Medical Subject Headings terms and free-text keywords, including, but not limited to: "Doping in Sports" AND (transdermal OR percutaneous OR transcutaneous OR dermal OR skin). For literature on the transdermal uptake of doping-relevant substances the identical search terms were combined individually with all substances (including name variants) on the WADA prohibited list, with the exception of Glucocorticoids. Within the individual substance groups on the Prohibited List, only a few relevant representatives were considered for this review, due to the larger number of available studies. In general, only peer-reviewed journal articles published in English were included and where available, human clinical studies were prioritized over animal or in vitro experiments. To ensure sufficiency, articles were critically appraised for relevance and methodological quality. Redundant or irrelevant articles were systematically excluded through a stepwise screening process involving title, abstract, and full-text review.

Biopharmaceutical aspects of transdermal drug delivery

Human skin structure and function

Covering the entire surface of the human body, the skin represents the largest organ and fulfills a multitude of vital biological functions. It acts as a barrier against physical, mechanical, or chemical damage, helps to regulate body temperature, prevents the excessive loss of water and electrolytes, and protects the human body from ultraviolet radiation from the sun.[3,22,23] From top to bottom, the human skin is composed of three layers: the epidermis with its associated appendages, sweat and sebaceous glands, the dermis, and the hypodermis.[22], **Fig. 1** schematically illustrates the different layers of the human skin.

As the outermost layer, the epidermis has a thickness of 50–200 µm, depending on the area of skin, and renews itself constantly.[24,25] The major type of cells is keratinocytes, which undergo a specific differentiation process terminating in the formation of anucleate corneocytes, which are eventually shed from the skin surface (desquamation).[26,27] According to keratinocyte morphology, the epidermis is divided into four layers: basal layer (stratum basale), squamous layer (stratum spinosum), granular layer (stratum granulosum), and horny layer (stratum corneum). In the basal layer, the deepest layer, cells proliferate and differentiate into keratinocytes and migrate into the stratum spinosum, where the cornification of the keratinocytes is gradually initiated.[28] In the stratum granulosum, the cells contain granules, formed by the accumulation of keratohyalin, protein structures that contribute to building a three-dimensional network of keratin.[3] At the transition to the stratum corneum, the keratinocytes are converted into dead corneocytes. This involves the loss of cell organelles, while the cytoplasm of the corneocytes is increasingly accumulated with keratin filaments, which are vital for the barrier function of the skin.[23] As the most superficial layer of the epidermis, the stratum corneum provides mechanical protection for the deeper tissues and is also mostly responsible for preventing the loss of water and restricting the penetration of foreign substances.[23] The intercellular space between the corneocytes consists of large proportions of ceramides, sterols, sterol esters, and free fatty acids, structured in low-permeable lipid bilayers, alternating with watery channels.[29,30] The generally low water content in the stratum corneum further impedes the diffusion of chemicals and microbial contaminants.[3] A hydrolipidic film with a pH value of around 4.5 on the surface of the stratum corneum, referred to as the "acid mantle" of the skin, additionally supports the barrier function.[31]

Underlying the epidermis, 1–4 mm of the connective tissue- and collagen-rich dermis are found.[32] The superficial papillary layer of the dermis contains blood vessels, lymph vessels, epidermally derived appendages, and nerve endings, including sensory receptors.[22] The deeper reticular dermis consists largely of collagen fibers and is vital for the mechanical strength and elasticity of the skin. Macrophages, mast cells, and lymphocytes in the dermis are pivotal to the skin's immune function.[5]

The subcutaneous layer, the hypodermis, functions as a connective tissue between skin, muscle, and bones. It contains large

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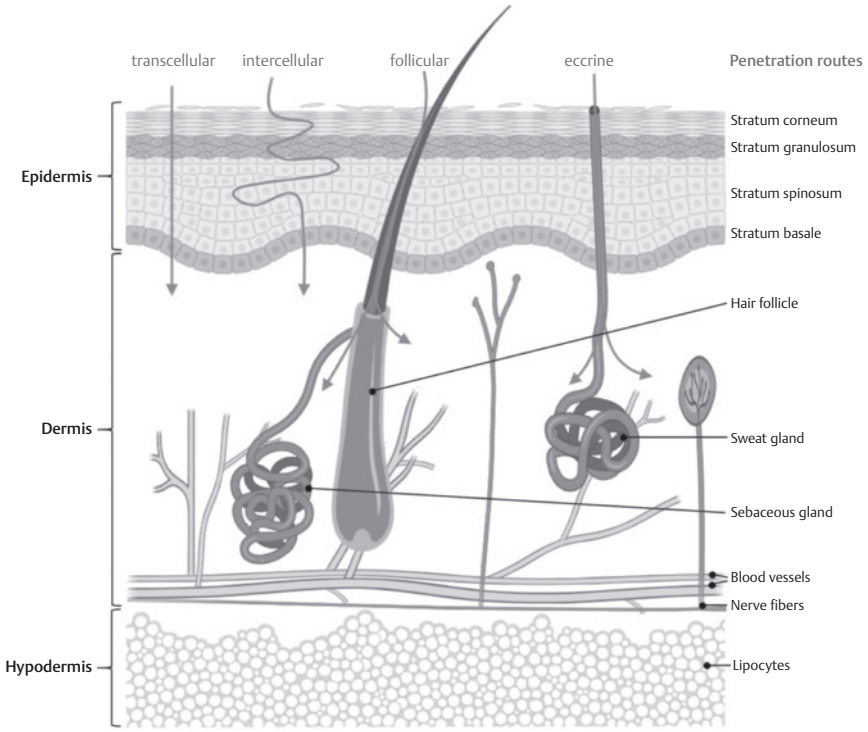

Fig. 1 Schematic illustration of the human skin and the major routes of drug penetration (transcellular, intercellular, follicular, eccrine), depicted as blue arrows. Created in BioRender. Krombholz (2025) BioRender.com/b42p470.

blood vessels, collagen, and lobules of fat cells (lipocytes), which play an important role in thermal insulation and protection from mechanical injuries.[5,23] This fatty tissue varies in thickness depending on the skin area and has become a prominent application site for the administration of peptide- and protein-derived drugs to overcome issues of poor bioavailability.[24,33]

Drug absorption via the skin

When applying drugs to the human skin, a mere superficial effect may be desired, e.g., in skin disinfection or insect protection. More commonly, a local effect is intended to treat skin conditions, usually localized in the epidermis or upper dermis. Transdermal drug delivery, however, refers to the systemic absorption of a drug after dermal application and represents an alternative to oral drug administration.[2] One benefit of the transdermal route is the avoidance of the gastrointestinal and hepatic first-pass effect and thus a possible improvement in bioavailability, e.g., of estrogens.[34] Furthermore, general variability in gastrointestinal absorption, associated with pH or food intake, can be circumvented.[29] The continuous drug release by transdermal delivery generally achieves stable blood levels. Thereby, spikes in plasma concentration that are associated with adverse drug reactions can be avoided, which is desirable, e.g., for opioid analgesics. Furthermore, reduced application intervals can overall improve compliance.[35–36]

For a drug to enter blood circulation, it must penetrate the different skin layers in sufficient amounts, thus overcoming the very obstacles our body has evolved to protect us from foreign substances. Essentially, there are two possible pathways: the transepidermal route and the route through the skin appendages, i.e., the sweat glands (transglandular) and hair follicles (transfollicular).[37] The major routes of transdermal drug penetration are depicted in **Fig. 1**. Presumably, more than one specific pathway is involved in the skin permeation of a particular drug. However, since pores only comprise an area of 0.1–1 % of the total skin surface, transglandular/transfollicular permeation is attributed to a subordinate role.[31,38,39] Drug molecules can permeate the epidermis through the cells (transcellular) or by passing between the corneocytes through the lipid matrix (intercellular), the latter being the dominant pathway for most low-molecular substances.[37] As this can essentially be seen as a passive diffusion process, the thickness of the diffusion layer, i.e., the skin, and the concentration gradient, i.e., the concentration of the drug, are decisive influencing factors.[29] Lipophilic properties are important to penetrate into the lipoidal layers, but very lipophilic components can also be retained in the stratum corneum as they hardly penetrate into the (more hydrophilic) layers of the viable epidermis.[40] In general, primarily nonionic drugs with a molecular weight < 500 Da, moderate lipophilicity (log*P* 1–3), and water solubility (> 1 mg/ml) appear to be capable of overcoming the epidermal barrier and to be absorbed in the dermal vasculature.[41] Hence, therapeutically relevant concentrations of the active substance are only obtained by a limited number of drugs. The difficulty of overcoming the skin barrier is reflected in the overall modest number of drugs actually approved for transdermal application, examples being nitroglycerin for the

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treatment of angina pectoris, fentanyl in pain management, or nicotine for smoking cessation.[42,43] In addition to the physicochemical properties and concentration of the drug substance, skin penetration can be influenced by a variety of other factors that will be discussed briefly.

Influencing factors

Skin physiology and condition

Skin area Depending on the anatomic site of application, a significant variability in skin permeation can be observed for some drugs.[44–47] Alongside other factors such as skin lipid composition, temperature, or the density of hair follicles, this can mainly be attributed to the varying thickness of the stratum corneum.[29] To a certain extent, data suggest enhanced absorption in areas of increased appendageal density and thinner stratum corneum (e.g., scrotum, axilla), whereas absorption from the trunk or back appears to be lower.[48] However, percutaneous absorption of some compounds does not seem to be affected by different application sites.[49–50]

Age Skin age may also influence transdermal absorption. The skin of newborns is generally thinner and has a different lipid composition, which makes it significantly more permeable than adult skin.[20] At the other side of the spectrum, aging skin is characterized by epidermal atrophy and friability, which can likewise enhance the permeability of the stratum corneum.[29] Paradoxically, however, percutaneous absorption of some compounds is even lower in elderly skin. This seems to primarily concern more hydrophilic substances, e.g., hydrocortisone, presumably due to the lower water content in the stratum corneum of the elderly. Hydrophobic substances (e.g., testosterone, estradiol) appear to be less affected.[51]

Sex In principle, differences can be observed between the skin anatomy of men and women, such as increased skin thickness or comparably larger pore sizes in men.[52] Nevertheless, studies with transdermal clonidine or nicotine concluded that, in practice, there is no difference in the permeability of male and female skin.[53,54] Presumably, the general variability of the skin/skin condition of individuals has a much greater impact on skin permeability than sex.[55]

Ethnicity Literature regarding interracial variations in transdermal drug absorption is conflicting. The major and established difference between skin of color and fair skin is the content and epidermal distribution of the cutaneous pigment melanin, as well as differences in hair and fibroblast structure.[56] Some studies suggest a thicker or more compact epidermis in subjects from African/African-Caribbean descent compared to subjects from Eurasian or far east-Asian descent.[57,58] However, small sample sizes and the omnipresent confounding factor of interindividual variability in many studies hinder the statistically significant determination of other structural differences.[56] Analogous to this, the literature is contradictory with regard to transdermal absorption of substances between different ethnic groups,[59,60] which is why no general statements can be made at this point about differences in skin absorption between fair skin or skin of color. Further research on this topic is warranted.

Skin conditions and disorders Damage to the stratum corneum severely impairs its function as a barrier. For instance, cold and dry environmental conditions can lead to a splitting of the horny layer or the use of high concentrations of soap and detergents can even cause a dissolution of the skin proteins and lipids.[61,62] Consequently, the permeation of substances through the skin can be increased substantially. Diseases of the skin, such as psoriasis or eczema, as well as sunburn also damage the skin barrier and thus increase permeability.[41] Conversely, enhanced hydration of the skin can also increase permeation rates. High humidity, skin moisturizer, or wetting the skin, e.g., while bathing, causes the outer layers of the stratum corneum to swell and makes them more permeable.[63] However, the extent to which transdermal absorption is influenced by the hydration status of the skin depends on the physicochemical properties of the penetrant.[29] An increase in skin temperature due to external conditions or fever causes an increased fluidity of the skin lipids and can therefore also improve the diffusion of substances through the stratum corneum.[55] Blood circulation and therefore drug clearance are also improved at higher temperatures, but as transportation from the dermis is not usually the rate-limiting step, this effect is considered negligible.[29]

Drug formulation

Drug modification The physicochemical characteristics of the penetrant are decisive for its penetration through the outer skin layers and absorption into the blood vessels of the dermis. The use of a prodrug, i.e., the insertion of a cleavable chemical group, can be a way to increase lipophilicity and improve the permeation of a substance through the skin.[1] Once absorbed, the substance is metabolized into its active form. Representative drugs that have been studied for this approach are naltrexone or morphine.[64,65] The formulation of ion–pair complexes can also be a strategy to enhance skin permeability, as has been shown for bisoprolol or escitalopram.[66,67] By adding an opposite ion species to the pharmaceutical formulation, a neutral paired compound with an improved partition coefficient is obtained.[5]

Type of vehicle Drugs for transdermal administration can be incorporated into gels, creams, ointments, or patches. In the latter, drug release can either be controlled by a membrane underneath the drug reservoir (reservoir type) or the compound can be homogeneously distributed in a polymer matrix (matrix type).[68] Most vehicles interact with the skin in some way and thus have an influence on the release and penetration of the active drug substance. The viscosity of gels, creams, and ointments can significantly influence the diffusion process of the drug substance to the skin surface.[29] Skin occlusion, e.g., by a patch, prevents transepidermal water loss and enhances the hydration of the stratum corneum.[5] Drug substances can also be enclosed in various types of nanovesicles (e.g., liposomes) to facilitate transdermal delivery, which represents a highly promising strategy to enable the transdermal delivery of a broad spectrum of drugs, including peptide- and protein-based therapeutics or even vaccines.[5,69]

Penetration enhancement The base of a transdermal formulation usually remains in the stratum corneum and is not able to penetrate the lipid bilayers.[68] However, there are a variety of substances that can significantly increase the absorption of drug substances through the skin. These are referred to as penetration enhancers, which work in different and sometimes multiple ways. Commonly used are sulfoxides (e.g., DMSO), alcohols (e.g., ethanol), azones

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(e.g., laurocapram), fatty acids (e.g., oleic acid), and fatty acid esters (e.g., isopropyl myristate).[2] Many of these compounds are capable of changing or even destroying the packing order of the lipid bilayers. By interacting with the polar head groups or the hydrocarbon chains of the skin lipids, they reduce the density of the intercellular matrix and allow drugs to penetrate more easily.[2,5] In addition to these classic chemical enhancers, the permeability of the stratum corneum can also be significantly increased by physical methods, using ultrasound (sonophoresis), electricity (iontophoresis, electroporation), microneedles, thermal ablation or microdermabrasion.[1] Although these various physical enhancers may be comparably costly and less convenient to apply, they may be more effective in delivering macromolecules and vaccines and, thus, have received growing attention in current research.[4,5,70]

Transdermal administration of substances included in the World Anti-Doping Agency's Prohibited List

Among the transdermal therapeutics approved by the U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA), and national medicine agencies, there are several substances prohibited in professional sports by WADA.[42,43,71] For many more doping-relevant substances, in vitro and/or in vivo data on transdermal bioavailability has been collected, and their pharmacological effects and metabolism following percutaneous absorption have been investigated. As this might impact their use for doping purposes and their detectability in doping control samples, research on the transdermal absorption of several representative compounds on the WADA Prohibited List has been reviewed and will be summarized and discussed in the following sections. An overview of selected studies on the transdermal absorption and/or metabolism of doping substances is provided in **Table 1**. The category of glucocorticoids is not included, since they are prohibited only when administered by injectable, oral, oromucosal, or rectal route.

Nonapproved substances

Within section S0 of the WADA Prohibited List, substances are included that are not addressed in other sections and are not currently approved by any governmental regulatory health authority for human use.[71] This also applies to 2,4-dinitrophenol (DNP), introduced as a weight loss drug in the early 1930s, but discontinued due to substantial side effects like cardiovascular diseases.[72,133] Liang et al. investigated a microneedle patch to deliver esterified DNP to mice. Their findings indicate that the transdermal patch could achieve localized delivery of DNP into the targeted fat tissue and reduce the accumulation in the heart, liver, and kidney, associated with a lower cardiovascular risk.[72] These are only preliminary results based on animal studies; however, a potentially reduced toxicity in humans by transdermal as opposed to oral administration could potentially increase the popularity of DNP for weight reduction in professional sports and might thus become relevant. The body protective compound (BPC) 157 is also currently not approved for clinical use but is the subject of various studies as it appears to promote the healing of various tissues.[134–136] It is also investigated for dermal application, primarily for the superficial treatment of skin wounds and burns.[134,137] Yet here too, transdermal use for doping purposes must be considered, as internet-based providers offer the prohibited substance in the form of transdermal patches, advertising them for the regeneration of muscles and tendons.

Anabolic agents

The skin permeability of various steroid hormones has been known for a long time and has since been used therapeutically, particularly for the administration of the AAS testosterone.[138] For the treatment of hypogonadism, testosterone was initially developed in the form of scrotal patches (e.g., Testoderm) but is now also available in nonscrotal patches (e.g., Androderm), gels (e.g., Testogel) and solutions for transdermal administration (e.g., Axiron).[6] From a therapeutic point of view, first-pass metabolism of the liver is avoided and a controlled release into the circulation can be achieved, whereby the circadian rhythm of testosterone secretion can be reproduced.[138] Transdermal testosterone has a bioavailability of 9–14 %, and constant plasma levels are achieved, a fact that is also highly relevant for sports drug testing programs as avoiding peak concentrations can be a possible strategy of dishonest athletes to lower the risk of AAFs in doping controls.[138,139] Apart from deliberate ingestion, in recent years, there have been several incidents of unintentional transdermal exposure to testosterone, including skin-to-skin transmission to infants in the course of paternal testosterone treatment.[140–142]

Structurally similar to testosterone, 19-nortestosterone (nandrolone) is equally capable of crossing the skin barrier and exhibiting androgenic effects.[73,74] In addition, the absorption of the steroid precursors 4-estrenediol and 4-estrenedione by human and equine skin, as well as subsequent biotransformation to unconjugated nandrolone, was demonstrated.[143,144] The serum concentrations of unconjugated nandrolone were found to be higher after transdermal than after oral administration of a nandrolone precursor.[144] In a study by Gessner et al., dermal exposure to oxandrolone, metandienone, clostebol, and dehydrochloromethyltestosterone (DHCMT) was investigated by administering 10 mg of each of the corresponding substances either at the back of the neck, on top of the hand, on top of the lower arm, or in the palm of the hand. All steroids were dissolved in DMSO as a penetration enhancer.[14] Urine samples were collected and analyzed by GC–MS/MS and LC–MS/MS in accordance with standard initial testing procedures employed in WADA-accredited antidoping laboratories. All substances (or their respective metabolites) could be rapidly detected and, in some cases, be observed over a prolonged period of time (up to 14 d). Different application sites were also examined to determine whether the administration of substances via a brief contact, such as in the form of sabotage, could result in an AAF. However, it was generally concluded that interindividual differences were most likely the major factor influencing the time of detectability, besides skin location and substance characteristics.[14]

Since the end of the twentieth century, selective androgen receptor modulators (SARMs) have been developed to treat conditions like muscle atrophy. SARMs are, in comparison to AAS, commonly associated with a more favorable profile of undesirable effects concerning e.g., gynecomastia or hypogonadism.[145] Nevertheless, acute liver toxicity and a reduction in high-density

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Table 1 Selected studies on the transdermal absorption of substances on the WADA Prohibited List

Class		Subclass	Substance(s)	Status	Delivery system	Design	Reference
S0	Nonapproved substances		2,4-Dinitrophenol		Esterification, MN	In vivo (mice)	72
S1	Anabolic agents	Anabolic androgenic steroids (AAS)	Nandrolon		Alcoholic solution	*Ex vivo* (human skin)	73
					Alcoholic solution	In vivo (rats, monkeys)	74
			Metandienone		DMSO	In vivo (humans)	14
			Oxandrolone		DMSO	In vivo (humans)	14
					Oil-in-water cream	*Ex vivo* (human skin)	75
			DHCMT		DMSO	In vivo (humans)	14
			Clostebol	Clinical approval	DMSO	In vivo (humans)	14
					Water-in-oil cream	In vivo (humans)	10
			Testosterone	Clinical approval	Scrotal Patch, matrix type	In vivo (humans)	76
					Patch, matrix type	In vivo (humans)	77
					Gel	In vivo (humans)	78
		Other anabolic agents	LY-305	Clinical Phase I	Gel	In vivo (rats)	79
					Gel	In vivo (humans)	80
S2	Peptide hormones, growth factors, related substances, and mimetics	Erythropoietins (EPO) and agents affecting erythropoiesis	EPO		Micropile-Arry	In vivo (dogs)	81
					MN-Patch	In vivo (rats)	82
		Peptide hormones and their releasing factors	hGH		MN, iontophoresis	In vitro (rat skin tissue)	83
					Hyaluronic acid conjugation	In vivo (mice)	84
					Laser microporation	In vitro (porcine skin)	85
			LHRH		MN	In vitro (porcine skin)	86
			Leuprorelin		Iontophoresis	In vitro (porcine skin)	87
					Iontophoresis	In vitro (human skin)	88
			Triptorelin		NP-MN	In vivo (rats)	89
					Iontophoresis	In vitro (rabbit/human skin)	90
			Nafarelin		Iontophoresis	In vitro (human epidermis)	91
			GHRP-5		CPP conjugation	In vitro (3D-skin model)	92
			GHRP-6		Microemulsion	In vivo (mice)	93
		Growth factors and growth factor modulators	des(1-3)-IGF-1		CPP conjugation	In vitro (3D-skin model)	92
			IGF-1		Microemulsion	In vivo (mice)	93
			Capromorelin		Solution	In vivo (humans)	94
S3	Beta-2-agonists		Tulobuterol	Clinical approval	Patch, reservoir type	In vivo (humans)	95
			Formoterol		Alcoholic solution	In vitro (rats)	96
			Salbutamol	Clinical Phase I	Patch, matrix type	In vivo (humans)	97
			Terbutaline		Gel, nanovesicles	In vivo (rats)	98

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Table 1 Continued

Class		Subclass	Substance(s)	Status	Delivery system	Design	Reference
S4	Hormone and metabolic modulators	Aromatase inhibitors	Anastrozole		Silica membrane	In vitro (porcine skin)	99
					Patch, matrix type	In vivo (beagle dogs)	100
			Exemestane		Nanoemulsion	In vivo (minipigs)	101
					Gel, nanovesicles	In vivo (rats)	102
			Formestane		Cream	In vivo (rats)	103
					Cream	In vivo (humans)	104
			Letrozole		Nanoemulsion	In vivo (minipigs)	101
		Antiestrogenic substances	Raloxifene		Nanovesicles	In vitro (rat skin)	105
					Nanovesicles	*Ex vivo* (rat skin)	106
			Tamoxifen		Nanovesicles	In vivo (mice)	107
					Alcoholic solution	In vitro (porcine epidermis)	108
		Metabolic modulators	Insulin	Clinical Phase II	MN	In vivo (humans)	109
				Clinical Phase I	Jet injector	In vivo (humans)	110
				Clinical Phase II	Patch, microporation	In vivo (humans)	111
S5	Diuretics and masking agents		Hydrochlorothiazide	Clinical Phase I	Patch, matrix type	In vivo (humans)	112
					Patch, matrix type	In vitro (rat skin)	113
			Indapamide		Patch, membrane type	In vitro (rabbit skin)	114
			Furosemide		Cream, nanovesicles	In vivo (cats)	115
S6	Stimulants	Nonspecified stimulants	Amfetamine	Clinical approval	Patch, matrix type	In vitro (humans	116
		Specified stimulants	Methylphenidate	Clinical approval	Patch, matrix type	In vivo (humans)	117
					Patch, matrix type	In vivo (humans)	118
			Selegiline	Clinical approval	Patch, matrix type	In vivo (humans)	119
S7	Narcotics		Buprenorphine	Clinical approval	Patch, matrix type	In vivo (humans)	120
			Fentanyl	Clinical approval	Patch, reservoir type	In vivo (humans)	121
					Patch, matrix type	In vivo (humans)	122
			Oxycodon	Clinical Phase II	Patch, matrix type	In vivo (humans)	123
			Tramadol		Gel, nanovesicles	In vivo (rats)	124
					Gel	In vivo (cats)	125
S8	Cannabinoids		Acidic cannabinoids		Microemulsion	In vitro (synthetic membrane)	126
			Δ^8-THC		Patch, reservoir type	In vivo (guinea pigs)	127
P1	Beta-Blockers		Bisoprolol	Clinical approval	Patch	In vivo (humans)	128
					Patch	In vivo (humans)	129
			Propanolol		Alcoholic solution	In vitro (porcine skin)	130
			Carvedilol		Gel, nanovesicles	In vivo (rats)	131
					Patch, reservoir type	In vitro (guinea pig skin)	132

Abbreviations: CPPs, cell-penetrating peptides; MN, microneedles; NPs, nanoparticles.

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lipoprotein cholesterol have been observed for many SARMs and a strategy to overcome these side effects appears to be the development of transdermally available SARMs.[79,80,146] Evidence of safety and tolerability of the transdermal SARM LY-305 could be provided in a phase-I study, and the substance proved to be skin-permeable in sufficient quantities to show potent muscle effects.[80] Given that SARMs have been detected in routine doping controls with increasing frequency in the past few years, the transdermal application of LY-305 and other SARMs must be considered in the context of antidoping efforts, especially since creams and transdermal liquids containing SARMs like LGD-4033 or S-23 are already available via Internet-based providers.[147]

Peptide hormones, growth factors, and related substances

While the transdermal application of anabolic agents offers the possibility of avoiding undesirable side effects, the administration of therapeutic proteins/peptides via the transdermal route focuses on another aspect: since peptides and proteins are challenging candidates for oral drug delivery (poor epithelial permeability, enzymatic and chemical degradation in the gastrointestinal tract), they require parenteral administration. This is generally associated with painful injections and discomfort for patients, making transdermal delivery a viable, more convenient alternative.[87] In principle, given their size and hydrophilicity, peptides and proteins are not capable of overcoming the epidermal barrier.[148] However, great efforts in research and development in the past few years have shown that the transdermal route can be made accessible to protein- and peptide-based drugs, some of which are banned in professional sports by WADA regulations. In studies on rats and dogs by Peters et al. and Ito et al., the 34-kDa glycoprotein erythropoietin (EPO) has been successfully administered percutaneously using EPO-coated microneedle patches.[81,82] Similarly, human growth hormone (hGH, 22 kDa) has been effectively administered transdermally in animal studies by microneedles,[83] conjugation to hyaluronic acid[84] or with the aid of laser-assisted microporation,[85] indicating that these strategies could be applied to humans at some point in the future. In an effort to achieve continuous drug delivery, several studies have also been performed on the transdermal delivery of luteinizing hormone-releasing hormone (LHRH) and its analogs used for the treatment of certain cancer forms and in in vitro fertilization.[86,87,89,90,149] While two of the commercially available LHRH receptor agonists, buserelin and nafarelin, are already administered intranasally, transdermal delivery is being investigated as a further alternative to intramuscular/subcutaneous injection to make chronic administration of these therapeutics more convenient.[149] Hence, various in vitro experiments have been conducted on leuprorelin, nafarelin, and triptorelin, where the utilization of electric current in the form of iontophoresis was particularly successful in enhancing skin permeation.[87,88,90,91] Besides iontophoresis, the permeation enhancement using cell-penetrating peptides could be demonstrated on the growth-promoting peptides GHRP-5 and des(1-3)IGF-1,[92] while the percutaneous permeation of IGF-1 and GHRP-6 was achieved using water-in-oil microemulsions.[93] The potential for very small peptide-like structures to passively diffuse through the skin was recently demonstrated in a study by Sobolevsky et al.[94] Following an AAF argued to result from cutaneous exposure to a veterinary preparation containing the growth hormone secretagogue capromorelin, controlled excretion experiments were carried out. Although absorption through the skin required significantly higher doses and produced a lower response compared to oral ingestion, capromorelin was detected in the study participants' urine samples for at least 48 hours after exposure. Beyond the scientific studies discussed, it should be mentioned that internet-based providers advertise transdermal patches containing the synthetic peptide TB-500. The clinical relevance appears to be modest, but speculations regarding its use for performance-enhancing purposes have been raised.[150–151]

Beta-2-agonists

Beta-2-agonists are generally administered by inhalation to promote local bronchodilatation in the treatment of bronchial asthma and chronic obstructive pulmonary disorder (COPD).[152] However, with Hokunalin tape, there is also a transdermal therapeutic system containing the beta-2-agonist tulobuterol commercially available.[153] The product is considered beneficial for asthma patients, particularly to prevent asthma attacks in the morning owing to the continuous release of the drug, even at night.[95] In addition, there are a number of research projects investigating the possibility of transdermal administration of formoterol, salbutamol, or terbutaline to reduce the frequency of use for patients through the sustained release of the active ingredient.[97,98,154] While performance-enhancing and anabolic effects of systemically administered beta-2-agonists are perceived, there is an ongoing debate about the extent to which inhaled beta-2-agonists can provide a competitive advantage in certain sports.[155–157] This debate is driven by the high prevalence of asthma and the common use of inhaled beta-2-agonists among endurance athletes.[158,159] Transdermal administration is not yet part of this debate but could become more relevant in the future. Further research may be required to determine the systemic concentrations in humans after transdermal application and whether performance-enhancing effects could be observed in comparison to other routes of administration.

Hormones and metabolic modulators

Within the group of hormones and metabolic modulators, the percutaneous absorption of several substances is being clinically investigated. These include aromatase inhibitors, e.g., anastrozole,[99,100] letrozole,[101] exemestane,[101,102] or formestane.[103] Here, the primary therapeutic aim is to reduce side effects related to the oral administration of these drugs, such as hot flashes, bone fractures, or myalgia, while maintaining the substances' ability to reduce breast cancer-associated mortality.[100,101,103] Several animal studies have demonstrated the feasibility of transdermal delivery of aromatase inhibitors.[99–104] Furthermore, the availability of topical gels containing formestane on the black market underlines the relevance of transdermal administration of these substances for doping purposes, too. Similarly, for the antiestrogenic substances raloxifene and tamoxifen transdermal drug delivery employing nanolipid carriers or chemical enhancers has been successfully accomplished in animal studies, aiming to overcome their poor oral bioavailability and to reduce side effects.[105–108] Various types of transdermal delivery methods have also been researched for the

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endogenous peptide hormone insulin, essential for the management of diabetes.[160–162] Methods that have already undergone clinical trials are microporation, microneedle-based insulin delivery, and transdermal insulin jet injectors.[109–111] Although none of the evaluated methods have clinical approval yet, transdermal insulin administration could become important in the future, to offer patients an alternative to subcutaneous self-injections, which are often perceived as uncomfortable. Accordingly, this route of insulin administration could also become relevant for the misuse of its anabolic effects for doping purposes.

Diuretics and masking agents

Diuretics are prohibited in professional sports, as they can affect the excretion of other substances and hence their detectability.[163] The transdermal administration of, e.g., hydrochlorothiazide or indapamide, for therapeutic purposes in blood pressure management, is under research as it potentially allows for reduced application intervals and minimized side effects through continuous drug release.[112–114] The use of compounded transdermal furosemide in cats with congestive heart failure appears to be a convenient way to apply the diuretic for both cat and owner, although a study by Sleeper et al. showed that skin absorption of furosemide in cats was poor compared to oral and intravenous drug administration.[115] For the transdermal administration to be effective, formulations thus need to be relatively concentrated, and a risk of exposure to the applying person has to be considered, particularly with regard to professional athletes.

Stimulants

The two substances responsible for most AAFs within the drug class of stimulants have in the past few years been approved by the FDA as transdermal patches for the treatment of attention-deficit hyperactivity disorder: amphetamine (Xelstrym) and methylphenidate (Daytrana).[116, 147, 164, 165] Here, the therapeutic benefits are sustained blood levels associated with less frequent dosing, as well as the possibility to uniquely adjust the duration of the drug's effect.[165] As can be expected, the plasma half-life of both drugs is significantly prolonged compared to oral immediate-release dosage forms.[164, 166] Interestingly, methylphenidate shows differences in its metabolism, as it undergoes an extensive and stereospecific first-pass effect when administered orally.[166] While *d-threo*-methylphenidate levels found in urine are about 10-fold higher than *l-threo*-methylphenidate, the difference in concentrations of both enantiomers is significantly lower with transdermal application.[117, 167] In animal studies, this also had an effect on the circulating amounts of the metabolite *d-* or *l-threo*-ritalinic acid.[166] Another stimulant available as a transdermal patch in the United States is the monoaminoxidase inhibitor selegiline, approved for the treatment of major depressive disorders.[119, 168]

Narcotics

Opioid patches are amongst the most prominent examples of transdermal therapeutic systems. Transdermal buprenorphine and fentanyl became available in the 1990s; both are strong opioids possessing physicochemical properties particularly favorable for skin absorption (low molecular mass, solubility in both fat and water, and high potency).[120, 169, 170] The possibility of transdermal administration is also being investigated for other opioids, such as oxycodone or tramadol, but is proving much more difficult.[123, 124] With regard to the pharmacokinetics, transdermal fentanyl has a plasma half-life of about 17 hours, which is attributed to a relatively fast diffusion into the lipophilic epidermal tissue, but subsequent slow transport through the water-rich dermal tissue into systemic circulation.[122, 171] For this reason, transdermal patches of fentanyl and buprenorphine are primarily intended for the treatment of chronic pain.[170, 172] In view of antidoping regulations, which prohibit the use of opioids in competition, a prolonged washout phase compared to oral administration must be considered. Additionally, as some opioids (e.g., tramadol) are compounded as transdermal gels for the treatment of pain in cats, a risk of exposure for athletes during the application of respective drugs is conceivable.[125]

Cannabinoids

Analogous to stimulants and narcotics, the use of cannabinoids is prohibited in competition by current antidoping regulations, except for cannabidiol (CBD).[71] To date, the transdermal administration of cannabis products or single cannabinoids appears to be of minor relevance compared to oral consumption or inhalation. With log*P* values of 5–7 and low water solubility, skin permeation of cannabinoids is considered challenging.[173, 174] Nevertheless, efforts are being made to improve the percutaneous delivery of cannabinoids by chemical permeation enhancers, microemulsion, or (nano)encapsulation.[126, 127, 175] In a study by Valiveti et al., a transdermal patch with Δ^8-tetrahydrocannabinol (THC) in propylene glycol/water/ethanol (1:1:1) was able to reach steady-state plasma concentrations 1.4 hours postadministration, lasting for at least 48 hours.[127] This indicates a significant accumulation of THC in the epidermis and a slow, sustained transport in the blood stream. In the context of therapeutic options, e.g., for cancer patients this can be considered beneficial, hence a positive trend for transdermal cannabis products can be expected in the future.[174] With regard to potential doping offenses, however, prolonged washout phases for transdermal in comparison to oral or inhaled cannabinoids must be assumed.

Beta-blocker

According to the WADA Prohibited List, beta-blockers are only prohibited in certain sports that can benefit from their ability to reduce the heart rate, e.g., archery or golf.[71] In general, beta-blockers are taken orally and many of them are not well suited for transdermal administration due to their hydrophilic properties.[176] Nonetheless, a considerable amount of work has been devoted to improving their skin permeation, and with Bisono Tape (Toa Eiyo Ltd., Japan), a transdermal delivery system for bisoprolol has recently been approved in Japan for the treatment of atrial fibrillation.[128, 129, 177] The formulation of transdermal gels and matrix-based patches has also been evaluated in vitro for other beta-blockers, e.g., propranolol or carvedilol.[130–132] In general, there appears to be a trend toward providing alternatives to the oral administration of beta-blockers to reduce application intervals for patients through continuous drug release via the skin.[176] Consequently, consideration should also be given to the abuse of such transdermal delivery systems for performance-enhancing purposes.

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Implications for sports drug testing

Skin as a potential, suspected, and/or proven contamination site

Pharmacologically, the skin poses an interesting way to administer substances more easily, in a more controlled manner, or with fewer side effects, but at the same time, it offers a large surface through which exogenous substances can inadvertently enter the human body.[2] This route of drug exposure must be considered in (forensic-)toxicological evaluations and, albeit under different considerations, is becoming increasingly relevant in the context of athlete doping controls.[15, 178] The sensitivity of analytical methods in doping control laboratories nowadays allows for the detection of the smallest amounts of prohibited substances in an athlete's blood or urine sample. Evidently, this is a significant achievement for antidoping efforts in general but simultaneously increases the risk of an AAF that is not the result of a doping scenario but the consequence of unintentional exposure to a prohibited substance.[16, 19] Possible situations leading to the aforementioned findings include, e.g., the ingestion of contaminated nutritional supplements, the consumption of food containing prohibited substances, or the transfer of analytes via intimate contact.[179–181] According to Daughton, either residues remaining from topically administered drugs or systemic residues excreted in sweat can be transferred to another person via skin-to-skin contact and enter the body through the transdermal route.[178] The latter scenario has recently been discussed in a case involving the SARM ostarine.[11] Following two consecutive AAFs with urinary levels of ostarine < 60 pg/ml, the athlete (A) argued he had worn the hamstring sleeves of another athlete (B) in the previous weeks, who had been abusing the anabolic agent before. Consequently, ostarine was believed to have been absorbed into the sleeves via sweat and transported through the skin of athlete A via a medical cream he was applying at the time. In subsequent investigations, several pieces of the hamstring sleeves were tested positive for ostarine, as well as hair and nail samples of abusing athlete B, while corresponding samples from athlete A were found negative. Therefore, sports authorities agreed that the AAF most likely resulted from contamination of the athlete by the hamstring sleeves. It has been shown that ostarine can be detected for up to 9 days after oral ingestion of only 1 µg.[182] However, data on the detectability of transdermal ostarine are not available yet and could be of additional use for the evaluation of cases as the one described here.

As previously stated for testosterone, the potential transfer of residues of topically applied drugs from one person to another is a well-known issue.[140] Recently, there has been an increasing number of cases, where athletes tested positive for the anabolic steroid clostebol, and transdermal contamination with topical formulations of clostebol was the most frequently mentioned argument and explanation.[13] To assess the plausibility of these contamination scenarios, de la Torre et al. conducted a controlled administration study and could demonstrate the detectability of the main metabolite (M1) as glucuronide conjugate after topical application of 5 mg clostebol acetate, even after only transient contact with the application area, mimicking the contamination of individuals applying the cream to others.[10] In addition to clostebol, Gessner et al. also administered oxandrolone, metandienone, and DHCMT to healthy volunteers via the skin.[14] Here, 10 mg of each substance was dissolved in only 15 µl of DMSO to assess, whether even a small drop on the skin was sufficient to detect the substance or respective metabolites in the participants' urine samples. To imitate contamination, the substances were applied on the palm or back of the hand, simulating a handshake. In addition, a brief embrace or friendly slap was investigated by applying the substances on the neck or on top of the lower arm. Apart from a large variability in the detection windows, all substances and/or their specific metabolites could be detected rapidly and in some cases for up to 14 days. Current research in our laboratory on the transdermal absorption of SARMs has shown that as little as 10 µg, applied to the forearm in a DMSO-containing cream, produced detectable amounts of LGD-4033 and S-23 in subsequently collected urine samples (data not published). Particularly noteworthy were the long detection windows following transdermal administration: a carboxyl-metabolite of LGD-4033 was detected for up to 13 days, and S-23 was even detectable for up to 16 days, demonstrating that modern test methods provide the sensitivity to produce AAFs for weeks at exposure levels of as little as µg amounts. Consequently, both athletes and their teams need to be aware of the risk of contamination with anabolic agents via various routes, including the skin, to avoid unintentional ADRV. The same can be concluded for the dipeptide capromorelin, which is approved by the FDA and EMA as an appetite stimulant for cats (Elura) and dogs (Entyce).[183, 184] In an administration study by Sobolevsky et al., the transdermal absorption of 30 mg capromorelin (corresponding to approximately 1 ml of Entyce) was investigated in three volunteers.[94] With concentrations ranging between 0.02 and 0.7 ng/ml, the substance was detectable in the collected urine samples for at least 48 hours. Thus, administering the drug to a pet appears to pose a significant risk for athletes to be contaminated in the process.[94, 185]

The overall issue of AAFs that are not the result of doping, but instead result from unintentional exposure of athletes to doping substances in their environment, is a concern for antidoping organizations, laboratories, as well as athletes.[185] Under global antidoping rules, athletes are held responsible for identifying the source of the prohibited substance in their sample and proving their innocence.[18] Piper et al. investigated the detection of DMSO as a common permeation enhancer to verify potential transdermal contamination; however, postapplication values hardly exceeded levels measured within a reference population.[186] As a pragmatic solution, minimum reporting levels have been proposed[16] and installed for selected analytes to account for known contaminants.[187] In addition, complementary measurements, such as hair and nail analysis, have been suggested and reported as additional evidence in assessing the plausibility of unintentional exposure in numerous instances.[11, 188, 189] Nevertheless, data on dose-dependent skin absorption rates of doping substances are of great value, alongside research into markers that might even allow to differentiate the route of drug absorption (oral/transdermal).

Influence of transdermal administration on the metabolic profile in doping controls

Besides monitoring the intact drug, doping control analyses frequently target metabolites of prohibited substances. With regard to anabolic steroids, the sensitive analysis of certain (long-term)

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metabolites of, e.g., metandienone or DHCMT has significantly improved the ability to uncover their misuse.[190–192] In addition, the metabolic profile can contribute to an assessment of the time or amount of the administered substance, help to assign the substance's origin (exogenous or endogenous), and potentially even delineate the route of drug administration.[193] Therefore, in the last decades, much effort has been put into the identification and characterization of metabolites of prohibited substances, often focusing on oral drug administrations. For testosterone, however, it is known that the formation of some metabolites varies depending on the route of administration (intramuscular/oral/transdermal).[7,194] Transdermally administered testosterone is converted into its 5α-reduced metabolites (androsterone, 5α-androstanediol) to a much larger extent than under conditions of endogenous biosynthesis or injected formulations, attributable to a high activity of the skin-borne 5α-reductase.[7] Hence, current research into testing and improving markers indicating the misuse of pseudoendogenous AAS is often applied to several routes of administration available for testosterone.[8,195,196] This has, for instance, contributed to an extension of the steroid profile to serum analysis, as it could be shown that serum testosterone and 4-androstenedienone are more sensitive markers to detect topical testosterone application than the steroid excretion pattern in urine.[197,198] The biotransformation of testosterone and several other AASs in human skin cells (human keratinocytes and human fibroblasts) was recently investigated by Liu et al.[199] Consistent with a high activity of 5α-reductase in human skin, only 5α-reduced metabolites were identified, such as 5α-dihydrotestosterone for testosterone and mestanolone for methyltestosterone. Several 5α-reduced metabolites were also found for clostebol and methylclostebol and could be identified by comparison to synthesized reference material. However, the main metabolite for clostebol formed after oral application (M1) exhibited only weak abundances here. For DHCMT and metandienone, no metabolites could be detected. Interestingly, Gessner et al. studied the detectability of several of these steroids and their metabolites in urine following transdermal administration to healthy volunteers.[14] Overall, it can be concluded from this study that not only the physicochemical properties and the application site influence the pharmacokinetics and biotransformation of transdermally administered substances. Presumably, several of the factors discussed in the section "Skin physiology and condition" (e.g., skin condition and age) add to a large variability between different individuals, which evidently complicates general conclusions. For oxandrolone, all investigated metabolites were detectable in the participants' urine samples, without any pronounced differences in the results following oral administration. Different from the experiments by Liu et al., the parent compound of metandienone was not detected in the participants' urine samples, however all of the examined metabolites were observed. The pattern was similar to an oral application, but a prolonged detectability of 6-OH-metandienone was considered potentially indicative of a transdermal administration. For DHCMT, the results suggested a prolonged detectability of DHCMT glucuronide (up to 14 d), as well as the absence of certain metabolites to indicate application via the skin. Clostebol was mainly detected in the form of its metabolite M1, corroborating the results of de la Torre et al. after topical application of a clostebol cream.[10] Additionally, de la Torre et al. identified other (less abundant) metabolites of clostebol, including the sulfo-conjugated metabolite M5. To potentially allow for discrimination between transdermal and oral clostebol administration, the ratio of M1 glucuronide and M5 sulfate was discussed as a preliminary marker in this study. Evaluating the utility of this marker warrants further investigations on the basis of a larger number of volunteers, and analytical means supporting decision-making processes in managing cases such as those of (alleged) clostebol contamination via the skin (vide supra) are particularly desirable. The metabolites detected in urine samples after transdermal AAS administration, as well as those obtained from the skin incubation experiments with several AAS, are listed in **Table 2**. Differences in the metabolic profile after transdermal and oral administration were also found for the antiestrogen formestane in a different study by de la Torre et al.[104] Not only was transdermal formestane detectable for a longer period of time, the concentrations of the metabolite 4-hydroxyandrosterone (4OH-A) were higher than those of 4-hydroxyepiandrosterone (4OH-EA) after transdermal application, when the opposite was true following oral administration. Thus, the ratio 4OH-EA/4OH-A could be utilized to elucidate the route of drug administration, if needed. This may be beneficial in the management of formestane AAFs in doping controls.

Another aspect that should be pointed out in this context is the deliberate choice of the transdermal route of drug administration, seemingly to undermine routine doping control analytical approaches. In a book on the systematic doping in Russia prior to and during the Olympic games in Sochi 2014, the author states that buccal or transdermal drug administration was "the only low-risk and hopefully efficient ways to administer steroids."[21] Presumably, the detection windows of buccally absorbed steroids are shorter compared to oral administration.[200] However, whether "the most [...] detectable, long-term metabolites would not emerge" following buccal or transdermal administration, as claimed in the book, is questionable or at least considered to be dose dependent.[21,200] Nevertheless, it must be taken into consideration that bypassing the hepatic first-pass effect as well as (possibly different) skin-derived metabolic reactions can influence the metabolite pattern of doping substances in urine and thus affect their detectability in doping controls. In addition to the investigations of cases of presumed contamination via the skin, this aspect further underlines the importance of research addressing the biotransformation of doping substances following transdermal administration.

Consequences and opportunities for antidoping research

Most substances on the WADA Prohibited List are banned, regardless of the route of administration.[71] Nevertheless, analytical strategies that allow to evaluate the route of drug delivery can be highly valuable, as they can help to clarify scenarios of unintentional (dermal) drug exposure.[10,19] In addition, transdermal drug delivery may produce ambiguous results in doping control analysis as the detectability of certain metabolites and/or metabolite ratios may differ in comparison to oral administration.[10,14,104] In this context, as outlined above, controlled human elimination studies are particularly valuable, but can be difficult due to ethical and safety reasons.[201] Within the scope of ethical deliberations, animal studies are another option. However, it has to be considered that the skin

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Table 2 Metabolites of different anabolic androgenic steroids detected after incubation with human skin cells and/or in urine samples following transdermal administration of the respective compounds

Design	Compound	Metabolite	Reference
In vitro (human keratinocytes/human dermal fibroblasts)	Testosterone	17β-hydroxy-5α-androstan-3-one (*5α-DHT*)	199
		5α-androstane-3α,17β-diol (*Adiol*)	
	Methyltestosterone	17α-methyl-17β-hydroxy-5α-androstan-3-one (*Mestanolone*)	
	Clostebol	4-chloroandrost-4-en-3α,17β-diol	
		4-chloroandrost-4-en-3ß,17β-diol	
		4α-chloro-5α-androstan-3α,17β-diol	
		4α-chloro-5α-androstan-3ß,17β-diol	
		4-chloro-3α-hydroxy-androst-4-en-17-one (*M1*)	
	Methylclostebol	4-chloro-17α-methylandrost-4-en-3α,17β-diol	
		4-chloro-17α-methylandrost-4-en-3β,17β-diol	
		4α-chloro-17α-methyl-5α-androstan-3α,17β-diol	
		4α-chloro-17α-methyl-5α-androstan-3β,17β-diol	
	Metandienone	None	
	DHCMT	None	
In vivo (healthy male volunteers, 100 mg testosterone gel/day, 6 weeks)	Testosterone	Androsterone (*And*)	7
		5α-androstan-3α,17β-diol (*Adiol*)	
In vivo (healthy male volunteers, 10 mg steroid dissolved in DMSO)	Oxandrolone	17α-Hydroxy-17β-methyl-2-oxa-5α-androstan-3-one (*EpiOx*)	14
		17,17-dimethyl-18-nor-2-oxa-5α-androsta-13-en-3-one (*NorOx*)	
		17α-hydroxymethyl-17β-methyl-18-nor-2-oxa-5α-androsta-13-en3-one (*epiNW*)	
		17β-hydroxymethyl-17α-methyl-18-nor-2-oxa-5α-androsta-13-en-3-one (*NW*)	
	Metandienone	6β-hydroxy-17β-hydroxy-17α-methyl-androsta-1,4-dien-3-one (*6-OH-Mtd*)	
		17α-methyl-5α-androstan-3α,17β-diol (*THMT*)	
		17β-methyl-5β-androst-1-ene-3α,17α-diol (*EMD*)	
		17,17-dimethyl-18-nor-5β-androsta-1,13-dien-3α-ol (*NorEMD*)	
		17β-hydroxymethyl-17α-methyl-18-norandrosta-1,4,13-trien-3-one (*NW*)	
		17α-hydroxymethyl-17β-methyl-18-norandrosta-1,4,13-trien-3-one (*epiNW*)	
		17α-hydroxy-17β-methyl-androsta-1,4-dien3-one (*epiMD*)	
	DHCMT	4-chloro-6β-hydroxy-17β-hydroxy,17α-methylandrosta-1,4-dien-3-one (*6-OH-DHCMT*)	
		4α-chloro-18-nor-17β-hydroxymethyl-17α-methyl-5β-androst-13-en-3α-ol (*M3*)	
	Clostebol	4-chloro-3α-hydroxy-androst-4-en-17-one (*M1*)	
In vivo (healthy male volunteer, 5 mg clostebol cream)	Clostebol	4-chloro-3α-hydroxy-androst-4-en-17-one (*M1*)	10
		4α-chloro-5α-androst-3α-ol-17-one (*M2*)	
		4α-chloro-5β-androst-3α-ol-17-one (*M3*)	
		4α-chloro-16α-hydroxy-5α-androst-3α-ol-17-one (*M4*)	
		4α-chloro-5α-androstan-3β-ol-17-one (*M5*)	

Abbreviation: DHCMT, dihydrochloromethyltestosterone.

of some animals, e.g., rodents, differs greatly from the human skin and might not be a suitable surrogate.[202, 203] Consequently, alternative methods are needed to investigate the transdermal bioavailability and biotransformation of substances relevant in the context of sports drug testing. Transdermal drug transport can be predicted based on mathematical models, that include physicochemical parameters of the drug, drug concentration, or thickness of the skin.[204, 205] However, although many of these models can be quite

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complex, they often oversimplify biological processes and cannot fully account for the considerable heterogeneity of the skin.[206] Experimental skin models include *ex vivo* human or animal skin (obtained from surgery or cadavers), artificial membranes, or reconstructed skin, mounted in a diffusion cell.[207] While many of these models can be useful to determine drug permeability, with regard to studying skin metabolism, the presence of viable epidermal tissue and, thus, fresh tissue is required.[208] The availability of ex vivo human skin can be limited, which is why animal skin is often used as a surrogate. Here, porcine skin is preferred as it is remarkably similar to human skin in terms of structure, thickness, and hair-follicle density.[209, 210] In vitro experiments using human skin cells, such as keratinocytes or fibroblasts, can be a more convenient way to use human material, as demonstrated by Liu et al. or Savill et al. for various anabolic steroids.[199, 211] However, skin-only models do not reflect postabsorption processes such as hepatic metabolism. This becomes evident by looking at the described differences between the in vitro results by Liu et al. and the urinary metabolites detected by Gessner et al. following transdermal administration of several AAS.[14, 199] The coculture of skin and liver spheroids on an organ-on-a-chip platform could be an interesting in vitro option to better reproduce the processes involved in the human body.[212, 213] Lastly, in addition to studies aiming to find specific metabolites or metabolite ratios, research into the analysis using a complementary matrix (blood, dried blood spots, hair, nails), as well as the detection of concomitant substances in administered creams or ointments could be of value in the future.

Conclusion

Investigations into the athletes' exposome, potential sources of contamination, and how to differentiate intentional doping from unintentional exposure to prohibited substances have become central aspects of antidoping research.[15] Several cases of presumed contamination in recent years have shown that the skin can be a relevant route of exposure, stressing the need to address skin absorption and the pharmacokinetics of doping-relevant substances administered via the skin. This requires a deeper understanding of the complexity and multilayered nature of human skin, that has been outlined in this review. Transdermal drug delivery is influenced by a multitude of factors, mostly by the physicochemical parameters of the substance and the drug concentration, but also by the site of application, age, or condition of the skin. All of these factors seem to interrelate, thus complicating the prediction of the influence of specific variables, and a high inter-individual variability has been observed in previous studies on the transdermal administration of prohibited substances.[14, 94, 104] Hence, addressing the question of the reliability of specific metabolites/metabolite ratios requires further data to support the differentiation between transdermal and oral administration. Hence, a sufficiently large number of volunteers should be included in corresponding studies to strengthen the validity of the obtained results.

Whilst several substances on the WADA Prohibited List are approved as transdermal preparations (e.g., testosterone or fentanyl), many more are under research to evaluate their skin absorption and find suitable methods to administer these substances via the skin. These investigations primarily have a pharmaceutical or clinical background, but they also have to be considered from an antidoping perspective. That is because the clinical motivations to administer substances transdermally, i.e., a more convenient way of application, bypassing first-pass metabolism or the reduction of side effects could also encourage the misuse of these drugs for doping purposes.

Further investigations into the detectability and metabolite profile of substances administered via the skin are warranted, more specifically in the case of anabolic steroids, where the current detection methods used in doping control laboratories strongly rely on the detection of specific (long-term) metabolites. Controlled human elimination studies, as demonstrated within this review, are particularly informative. Nevertheless, the application of appropriate in vitro methods, e.g., organ-on-a-chip can also be helpful to reflect the metabolism of doping-relevant substances following skin absorption. Whether in the sense of deliberate doping or in the case of unintentional exposure, the skin as a site of drug resorption will continue to play a role in the context of antidoping.

Author affiliations

1 Institute of Biochemistry, German Sport University Cologne, Cologne, Germany

2 Center for Preventive Doping Research – Institute of Biochemistry, German Sport University Cologne, Cologne, Germany

Statements and additional information

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